ECA1804 – Optoelectronics and Optical Communication

EXP NO :4- Measurement and Analysis of Dispersion Compensation in Optical Fiber using OptiSystem.

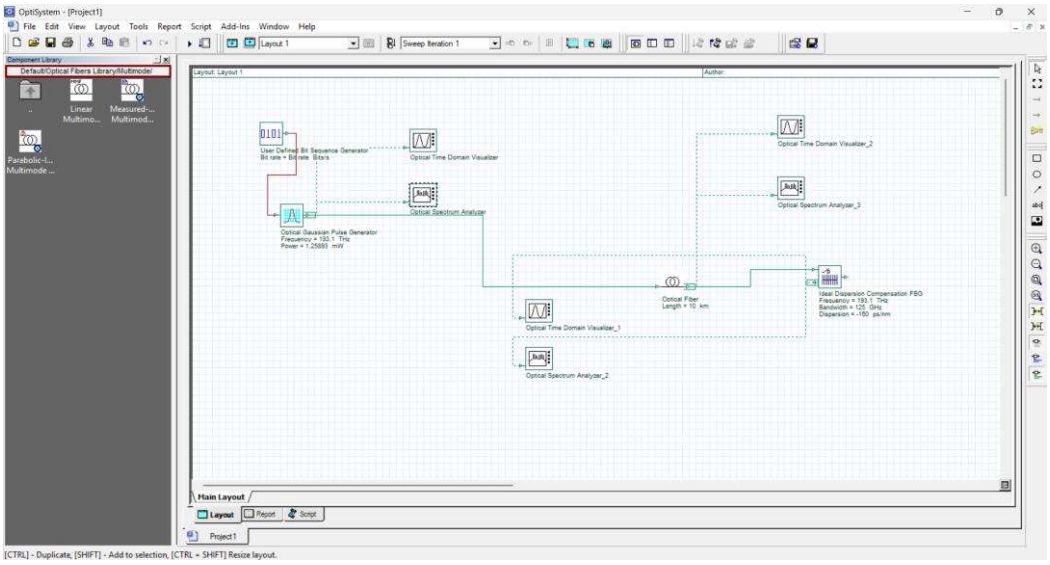
OBJECTIVES

1. To demonstrate dispersion compensation using the Ideal Dispersion Compensation component in OptiSystem.
2. To observe the broadening of a 12.5 ps optical Gaussian pulse after propagation through 10 km of single-mode fiber (SMF).
3. To compensate the accumulated dispersion of 160 ps/nm and verify recovery of the original pulse shape.

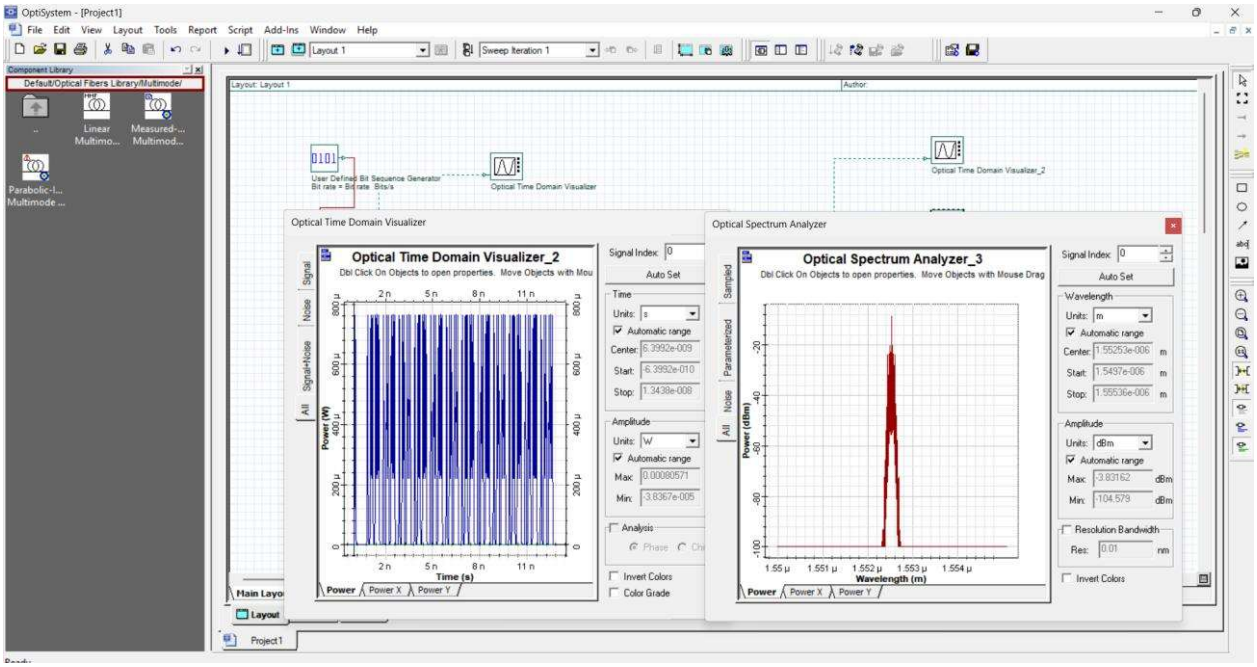
SOFTWARE / HARDWARE REQUIRED

- OptiSystem Software

Circuit Diagram:



Output:



Reflection:

Through this experiment, I learned how to measure and analyze dispersion compensation in optical fiber using OptiSystem. I understood how chromatic dispersion affects signal quality and causes pulse broadening. I observed how dispersion compensation techniques help reduce these effects and improve signal transmission. This experiment enhanced my understanding of optical communication systems and their performance.

Rubrics:

LABORATORY PERFORMANCE ASSESSMENT RUBRIC

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
Design the OptiSystem layout	Project layout	Builds the complete layout with correct components and connections independently	Minor connection errors corrected	Needs guidance for components or connections	Unable to build the layout	20
Configure pulse, SMF and compensator parameters	Parameter configuration	Sets all parameters (40 Gb/s, 12.5 ps, 193.1 THz, 10 km, −160 ps/nm, 125 GHz) correctly	Minor parameter errors	Needs guidance for several parameters	Unable to configure parameters	20
Observe pulse broadening in SMF	Pulse observation	Captures and interprets initial and broadened pulses correctly	Minor interpretation errors	Limited observation / needs help	Unable to obtain or interpret pulses	20
Verify compensation using ideal dispersion component	Compensation verification	Verifies recovery of the pulse and explains residual effects of settings	Mostly correct verification	Partial verification with assistance	Unable to verify compensation	20
Document findings	Report and conclusion	Complete report with justified conclusion	Minor omissions	Incomplete record / weak conclusion	Report / conclusion inadequate	20

Performance interpretation: Excellent = 85–100; Good = 70–84; Satisfactory = 50–69; Needs Improvement = below 50.

Result:

Dispersion compensation was successfully demonstrated using the Ideal Dispersion Compensation component in OptiSystem. A 12.5 ps Gaussian pulse broadened approximately four times after 10 km of SMF (accumulated dispersion 160 ps/nm), and sejing the compensator dispersion to –160 ps/nm restored the pulse, achieving an exact compensation of the accumulated dispersion.